

LIANGYU TANG

TARGET AREAS: Power System Planning and Operation | Grid Resilience and Vulnerability Analysis

Phone: +86 15982241128 Age: 28 email: 794282639@qq.com Expected Graduation: Jul 2027
Personal Website: <https://petert666.github.io/index.html>



PROFILE PhD candidate in Control Science and Engineering at the UESTC, conducting research on power-system resilience under extreme weather, power-network vulnerability identification.

PUBLICATIONS	RESEARCH PROJECTS	INTELLECTUAL PROPERTY	HONORS
6 first-author papers	3 State Grid projects	6 patent applications; 1 granted	SMC Scholarship, First Prize

EDUCATION

[Sep 2021-Present]	UESTC	Control Science and Engineering	PhD Candidate
Supervisor: Prof. Yang Han			
[Sep 2025-Jun 2026]	Politecnico di Torino	Department of Energy	Visiting PhD
Supervisor: Prof. Tao Huang			
[Sep 2016-Jun 2020]	Xihua University	Electrical Engineering	B.Eng.

RESEARCH INTERESTS

- **Power-System Resilience:** Resilience assessment, resilient scheduling, and power-supply assurance under extreme heat, disaster disturbances, and source-load uncertainty.
- **Vulnerability Analysis:** Identification of vulnerable buses, branches, and critical components using complex-network theory and power-flow mechanisms, accounting for both topology and operating state.
- **Optimal Scheduling:** Day-ahead, intraday, and real-time coordinated optimization of wind-PV-hydro-thermal-storage systems, including joint demand-response and energy-storage regulation.
- **Distribution-Network Fault Analysis:** Fault diagnosis and location, situational awareness, and critical-component risk assessment in distribution networks with high renewable penetration.

SELECTED PUBLICATIONS

1. **Liangyu Tang**, Yang Han, Amr S. Zalhaf, et al. Resilience enhancement of active distribution networks under extreme disaster scenarios: A comprehensive overview of fault location strategies. *Renewable and Sustainable Energy Reviews*, 2024. **First author | JCR Q1 | IF 18.0**
2. **Liangyu Tang**, Yang Han, Siyu Zhou, et al. Identification and vulnerability assessment of critical components in distribution networks under high penetration rate conditions. *Energy*, 2025. **First author | JCR Q1 | IF 10.1**
3. **Liangyu Tang**, Chaofeng Yan, Yuanhao Luo, et al. Resilient scheduling of wind-PV-hydro-thermal-storage multi-energy complementary systems under extreme heat. *Power System Technology*, 2026, 1-15 [in Chinese]. **First author | EI-indexed core journal**
4. **Liangyu Tang**, Yang Han, Chaofeng Yan, Congling Wang, Tao Huang. Resilience-oriented scheduling and evaluation of multi-energy complementary systems under extreme heat. *Electric Power Systems Research*, 2026. **First author | JCR Q2 | IF 4.3**
5. **Liangyu Tang**, Yang Han, Ping Yang, Congling Wang, Amr S. Zalhaf. A review of voltage sag control measures and equipment in power systems. *Energy Reports*, 2022. **First author | JCR Q2 | IF 6.6**
6. **Liangyu Tang**, Yang Han. Review of resilience enhancement strategies for active distribution networks under extreme disasters [in Chinese]. The 27th Annual Conference of the China Power Supply Society, 2024. **First author | Outstanding Presenter, Technical Session**

INVENTION PATENTS

1. Control strategy for an integrated hydro-wind-PV-storage multi-energy complementary system under extreme-heat conditions. **Granted | First-listed student inventor**
2. Risk identification model and system for distribution networks with high renewable penetration based on dynamic power-flow analysis. **Under substantive examination | First-listed student inventor**

3. Multi-time-scale resilient scheduling strategy for multi-energy complementary systems under extreme heat. **Under substantive examination | First-listed student inventor**
4. Power-system renormalization and equivalencing method for vulnerability-skeleton preservation. **Published application | First-listed student inventor**
5. Higher-order vulnerability identification method for power systems based on counterfactual propagation contributions. **Published application | First-listed student inventor**
6. Multi-stage power-grid reinforcement planning method considering security risks and weak grid links. **Published application | First-listed student inventor**

RESEARCH EXPERIENCE

- [Mar 2026-Present] Higher-Order Vulnerability Assessment of Transmission Lines via Contextual Propagation-Intensity Learning**
- **Research Focus:** Conditional propagation relationships and higher-order vulnerability in multi-line, multi-stage cascading failures in power systems.
 - **Key Outcome:** Completed the overall technical framework and part of the case simulations; developing propagation-source, susceptibility, relay, and composite higher-order vulnerability indices.
- [Oct 2025-Jan 2026] Vulnerability-Oriented Power-System Renormalization for Critical Skeleton Network Analysis**
- **Research Focus:** Cross-scale retention of critical skeleton branches during network reduction for rapid vulnerability screening of large-scale transmission grids.
 - **Key Outcome:** Across six benchmark systems, achieved bus compression ratios of 29.50%-56.12%, branch compression ratios of 20.41%-45.73%, and Top-10% vulnerable-branch hit rates of 94.74%-100%; manuscript under second-round review at *Reliability Engineering & System Safety* (JCR Q1).
- [Feb 2025-Sep 2025] Resilient Scheduling of Wind-PV-Hydro-Thermal-Storage Multi-Energy Complementary Systems under Extreme Heat**
- **Research Focus:** Developed unified time-varying constraint models for heat-induced changes in thermal-unit efficiency and capacity, PV conversion efficiency, line and transformer thermal ratings, and temperature-sensitive loads.
 - **Key Outcome:** Validated economic and resilience improvements on the IEEE 30-bus system and completed sensitivity analysis and bus-level interruption and restoration statistics; published in *Power System Technology*.
- [May 2024-Feb 2025] Critical Component Identification and Vulnerability Assessment in Distribution Networks with High Renewable Penetration**
- **Research Focus:** Unified vulnerability assessment of buses and branches across operating scenarios under distributed-generation uncertainty.
 - **Key Outcome:** Achieved a Pearson correlation coefficient above 0.97 between the proposed index and actual power-flow distribution; published in *Energy*.
- [Aug 2022-Sep 2023] Active Distribution-Network Fault Diagnosis, Location, and Extreme-Disaster Resilience Enhancement**
- **Research Focus:** Systematically reviewed feeder selection, fault-section location, precise fault location, and limited-micro-PMU placement methods, and analyzed diagnostic challenges introduced by distributed generation.
 - **Key Outcome:** Established a comprehensive research framework for fault location and resilience enhancement in active distribution networks; review published in *Renewable and Sustainable Energy Reviews*.

RESEARCH PROJECTS

- [Apr 2024-Dec 2025] SGCC Research on Integrated Hydro-Wind-PV-Storage Complementary Control Strategies for Multiple Application Scenarios**
- **Contribution:** Developed an optimal scheduling model for a hydro-wind-PV-storage multi-energy complementary system incorporating deep peak regulation of thermal units, energy-storage operation, and demand response.
- [Sep 2022-Sep 2023] SGCC Research and Application of Integrated End-of-Feeder Sensing, Intelligent Diagnosis, and Service Enhancement Technologies for Rural Low-Voltage Distribution Areas**
- **Contribution:** Cleaned electricity-consumption data from rural low-voltage distribution areas and developed distribution-network power-flow and theoretical line-loss calculation models.
- [May 2021-May 2022] SGCC Development of a Decision-Support Module for Voltage-Sag Mitigation Schemes for Critical Users in the Sichuan Power Grid**
- **Contribution:** Quantified voltage-sag tolerance and economic losses for critical users, developed a library of representative sensitive-equipment models, and designed decision logic for mitigation-scheme selection.